

Data quality and the BMI 30 artifact

Profiling, duplicate handling, sampling bias, and the discontinuity that identifies this dataset as simulated.

Provenance. The source file is the widely circulated Medical Cost Personal Datasets teaching set, which is simulated rather than drawn from real claims. The findings are real properties of the data and the methods are the ones that would be used on a live book, but no conclusion here describes an actual insurer or population.

Summary

The file is technically pristine and that is itself the most important observation about it. There are no missing values, no impossible entries, and near-perfect balance across sex and region. Real claims extracts do not look like this. Three specific properties identify the data as simulated, and one of them directly invalidates a finding that would otherwise look like the strongest result in the analysis.

Completeness and validity

Check	Result	Assessment
Rows	1,338 raw, 1,337 analysed	One duplicate pair removed
Missing values	0 across all columns	Implausibly clean for real data
Impossible values	None	No negative charges, no zero BMI
Age range	18 to 64	Within plausible bounds
Sex balance	676 male / 662 female	Near-perfect
Region balance	324 to 364 per region	Near-perfect
Charges skew	1.52	Right-skewed, as expected for claims

Issue 1 — the duplicate pair

Rows 195 and 581 are byte-identical: a 19-year-old male non-smoker with BMI 30.59, no children, in the northwest, charged \$1,639.56. Because the file carries no member identifier, a genuine coincidence and a duplication error are indistinguishable. One copy was removed. The effect on published figures is

negligible — total spend falls by 0.009% — but the removal is documented because silent row-dropping is how analyses become irreproducible.

Issue 2 — ages 18 and 19 are over-sampled

The file contains 69 members aged 18 and 68 aged 19, against a flat count of roughly 28 for every other age from 20 to 64. This is a sampling artifact rather than a population characteristic. Any unweighted trend across age inherits the bias, which is one reason the age analysis in this project reports the gap between smoker and non-smoker cost rather than the absolute level.

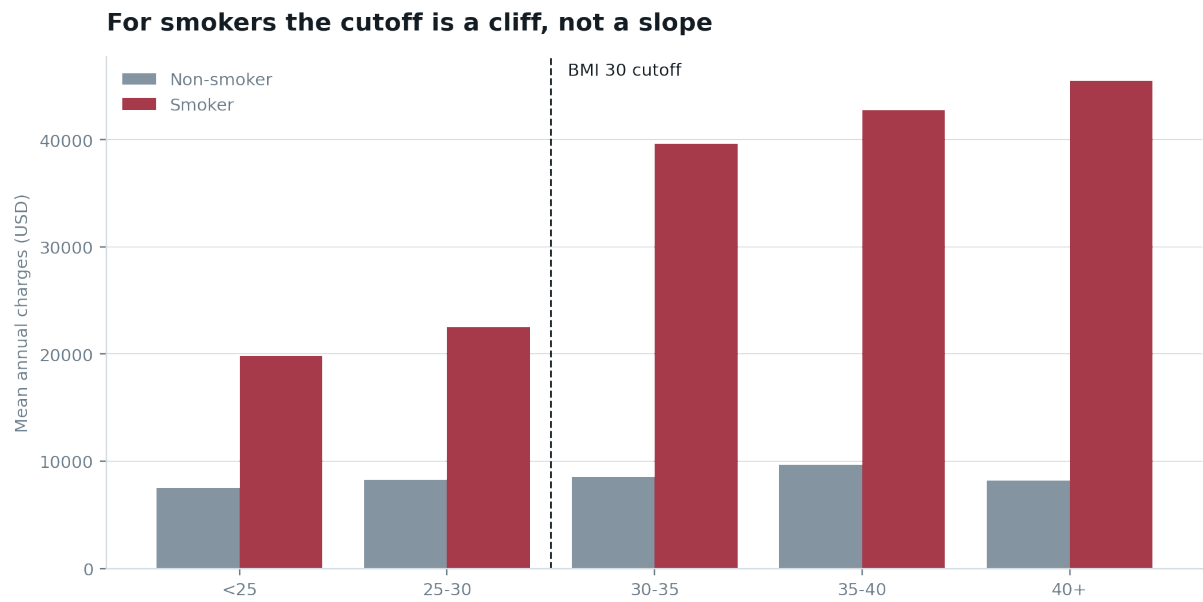
Issue 3 — the BMI 30 discontinuity

This is the finding that looks strongest and is least trustworthy, and it is worth setting out in full because a portfolio reader who recognises the dataset will be looking for it.

Fitting a simple linear trend within each side of the BMI 30 boundary, separately for smokers and non-smokers, gives the following.

Group	Slope below 30	Slope above 30	Step at the boundary	n below / above
Smokers	\$491 per BMI point	\$545 per point	\$20,195	129 / 145
Non-smokers	\$184 per BMI point	\$8 per point	\$879	502 / 561

Among smokers the within-band gradient is roughly \$491 per BMI point below the cutoff and \$545 above it — almost unchanged. Yet crossing the boundary itself is associated with \$20,195. A biological dose-response relationship does not behave like this. Metabolic risk rises continuously with adiposity; it does not wait at BMI 29.9 and then discharge twenty thousand dollars at 30.0.



The smoker series is essentially flat within each side of the cutoff and jumps across it.

Conclusion. The threshold is a rule in the data generator, not a property of human physiology. Any eligibility criterion, saving estimate or risk score built on BMI 30 in this dataset inherits the artifact. The weight-programme lever aimed at obese smokers is scored low-confidence for exactly this reason, despite showing the second largest theoretical ceiling in the analysis.

Issue 4 — a quarter of the variance is not in the file

An ordinary least squares model on all six available inputs, with categorical variables one-hot encoded and no interaction terms, reaches $R^2 = 0.75$. The residual is not randomly distributed. Eighty members sit more than \$10,000 above their predicted cost.

The clearest sub-group is 10 non-smokers billing over \$30,000 — 0.94% of all non-smokers, accounting for \$327k or 1.8% of the book. Their profiles are unremarkable: mostly aged 44 to 64, BMI between 24.7 and 37.7, spread across all four regions. Nothing in the available columns distinguishes them from members costing a fifth as much.

In a real book this is where a chronic condition registry, an inpatient admission flag or a claims history would sit. Its absence is the single most valuable thing this analysis reveals about the data collection, and it is why instrumenting the dataset is ranked third among the levers despite generating no direct saving of its own.

What the file cannot support

- **Any causal claim.** No time dimension, no baseline, no control.
- **Any trend or before-and-after comparison.** No dates.
- **Any per-member tracking.** No identifier.
- **Any cost driver analysis.** One aggregate charge, no claim lines, no diagnosis.
- **Any external benchmarking.** Charges are in an unstated price year.
- **Any conclusion about smoking intensity.** The field is a self-reported binary.

Ten questions covering what would need to be collected to answer the above are set out on the project dashboard, each with a metric, a measurement window and the decision it would settle.